

Project Scenario 1: Small Business Network

Part 1 – Basic Network Setup

Requirements

The company has:

- Management Department (VLAN 10)
- Sales Department (VLAN 20)
- IT Department (VLAN 30)

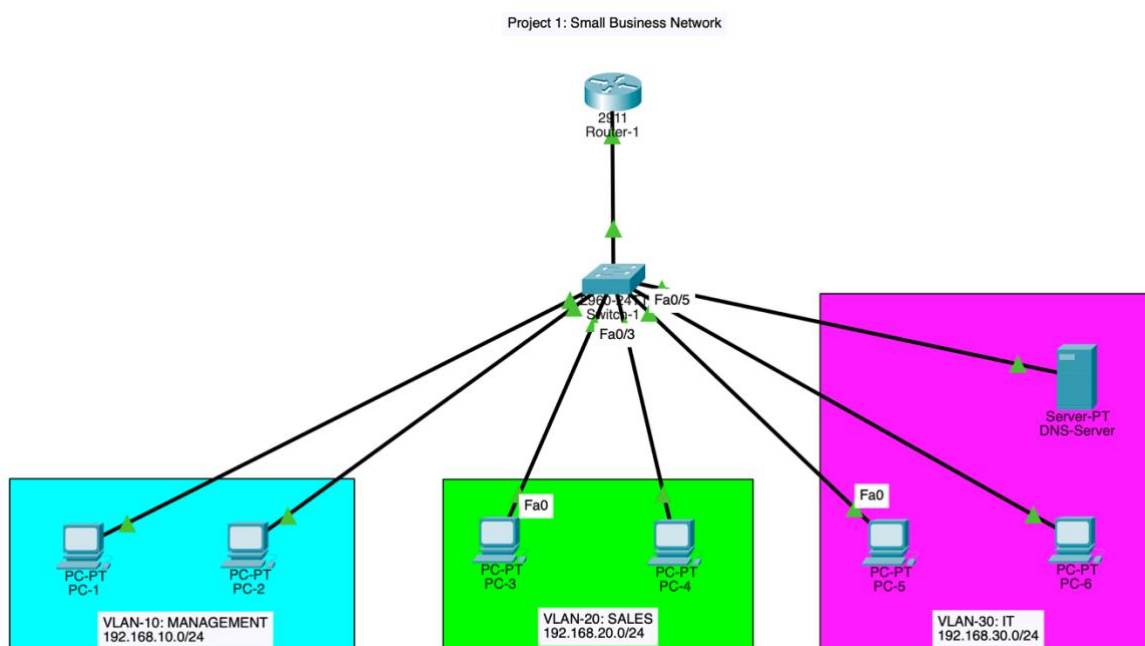
Each department should:

- Be separated using VLANs
- Communicate with other VLANs through Inter-VLAN Routing
- Receive IP addresses automatically via DHCP
- Use a DNS server for name resolution

1. Objective

To design and implement a small business network using VLANs, trunking, inter-VLAN routing, DHCP and DNS services within Cisco Packet Tracer.

2. Network Topology



3. IP Addressing Scheme

VLAN	Department	Network	Gateway
10	Management	192.168.10.0/24	192.168.10.1
20	Sales	192.168.20.0/24	192.168.20.1
30	IT	192.168.30.0/24	192.168.30.1

4. Switch Configuration

Create VLANS on Switch-1:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name MANAGEMENT
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name SALES
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#|
```

VLAN Brief:

```
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2

10   MANAGEMENT            active
20   SALES                  active
30   IT                     active
1002 fddi-default         active
1003 token-ring-default   active
1004 fddinet-default       active
1005 trnet-default         active
Switch#|
```

Assigning Ports to each VLAN:

Device	VLAN	Port
PC-1	10	Fa0/1
PC-2	10	Fa0/2
PC-3	20	Fa0/3

Device	VLAN	Port
PC-4	20	Fa0/4
PC-5	30	Fa0/5
PC-6	30	Fa0/6
DNS-Server	30	Fa0/7

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)# interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)# interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)# interface range fa0/5-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#
```

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# interface fa0/7
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#
```

Configure Trunking: The switch-to-router connection must carry all VLANs.

Since the router connected to interface g0/1:

```
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
```

Running-Config Verification:

```
Switch#show running-config
Building configuration...

Current configuration : 1409 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 30
 switchport mode access
!

interface GigabitEthernet0/1
 switchport mode trunk
!
interface GigabitEthernet0/2
!
```

5. Router Configuration

Configure Inter-VLAN Routing: This is Router-on-a-Stick.

Router interface:

```
Router>en
Router# conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown
```

Create sub interfaces:

```
Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#interface g0/0.30
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.30.1 255.255.255.0
Router(config-subif)#
```

Verify with interface brief:

```
Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	192.168.10.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.20.1	YES	manual	up	up
GigabitEthernet0/0.30	192.168.30.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

6. DHCP Configuration

The Router will provide DHCP. Exclude gateway addresses:

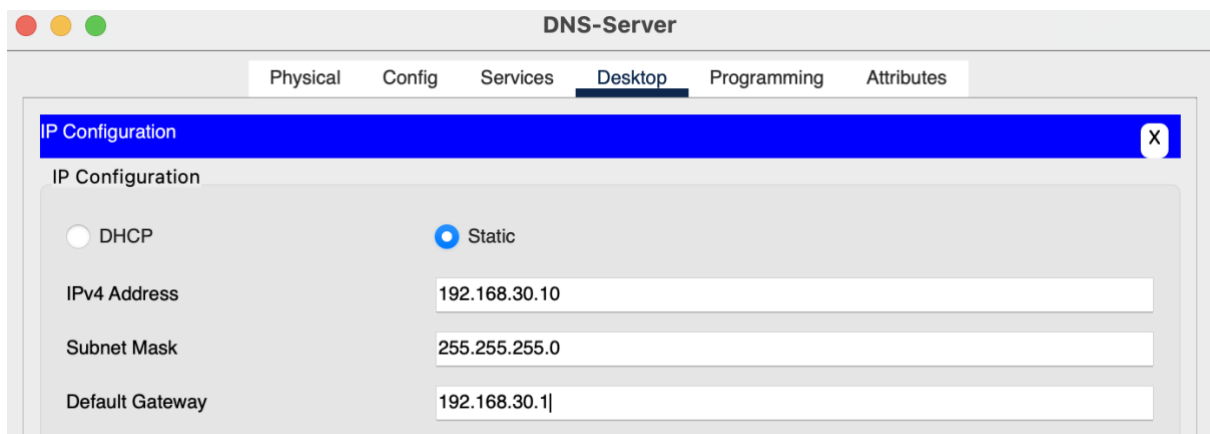
```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.10.1
Router(config)#ip dhcp excluded-address 192.168.20.1
Router(config)#ip dhcp excluded-address 192.168.30.1
Router(config)#
```

DHCP VLAN Pools:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool MANAGEMENT
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 192.168.30.10
Router(dhcp-config)#ip dhcp pool SALES
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 192.168.30.10
Router(dhcp-config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 192.168.30.10
Router(dhcp-config)#
```

7. DNS Configuration

Desktop IP Configuration:



The screenshot shows a window titled "DNS-Server" with a tabbed interface. The "Desktop" tab is selected. Below the tabs, there is a blue header bar labeled "IP Configuration" with a close button (X). The main area is titled "IP Configuration" and contains two radio buttons: "DHCP" (unselected) and "Static" (selected). Below these, there are three input fields: "IPv4 Address" with the value "192.168.30.10", "Subnet Mask" with the value "255.255.255.0", and "Default Gateway" with the value "192.168.30.1".

Service Records Config:

DNS-Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record**

Address

No.	Name	Type	Detail
0	company.local	A Record	192.168.30.10
1	fileserver.local	A Record	192.168.30.10

8. PC Configurations

All PC's are using DHCP for IP Configurations

PC-1

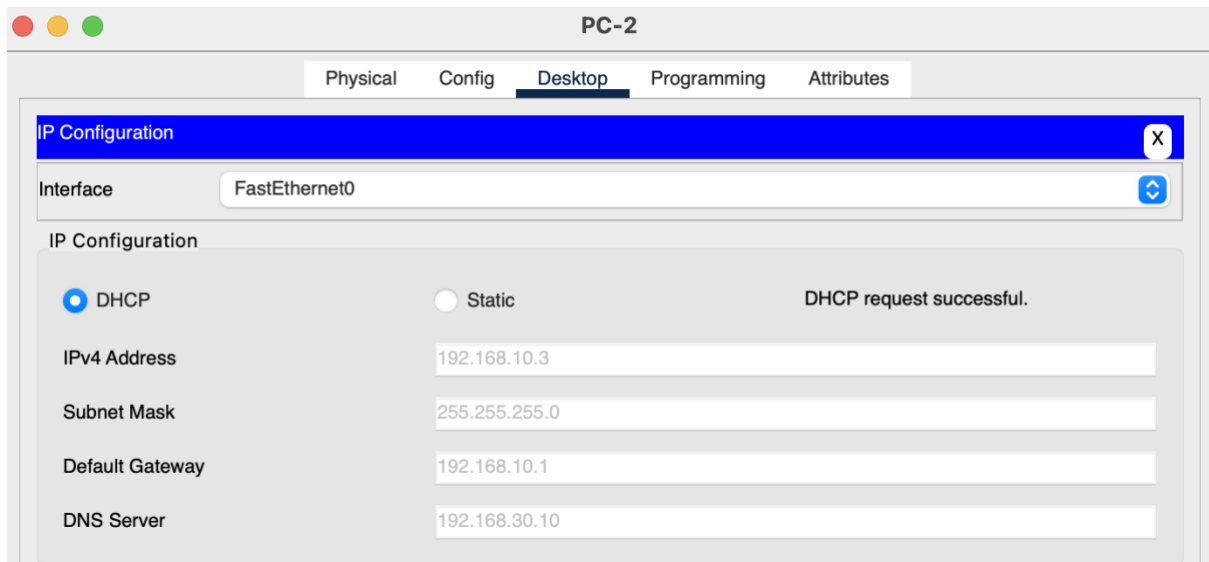
Physical Config **Desktop** Programming Attributes

IP Configuration

Interface **FastEthernet0**

IP Configuration

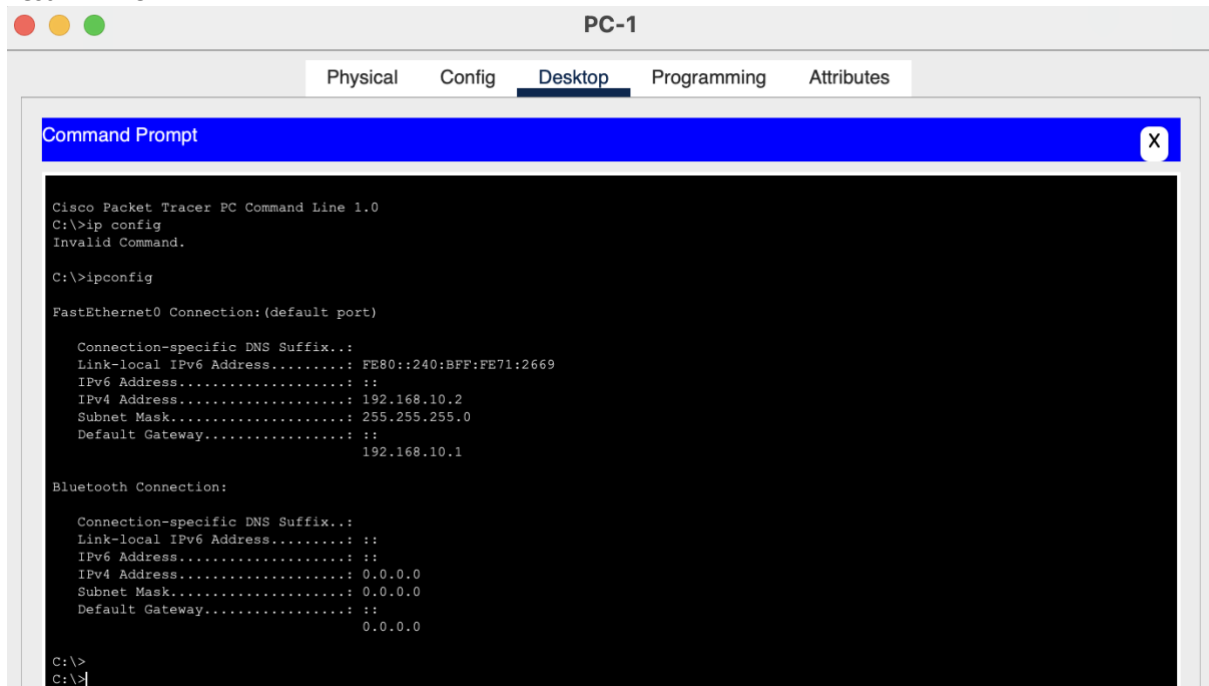
☒ DHCP ☐ Static Requesting IP Address



9. Testing Results

Include screenshots of:

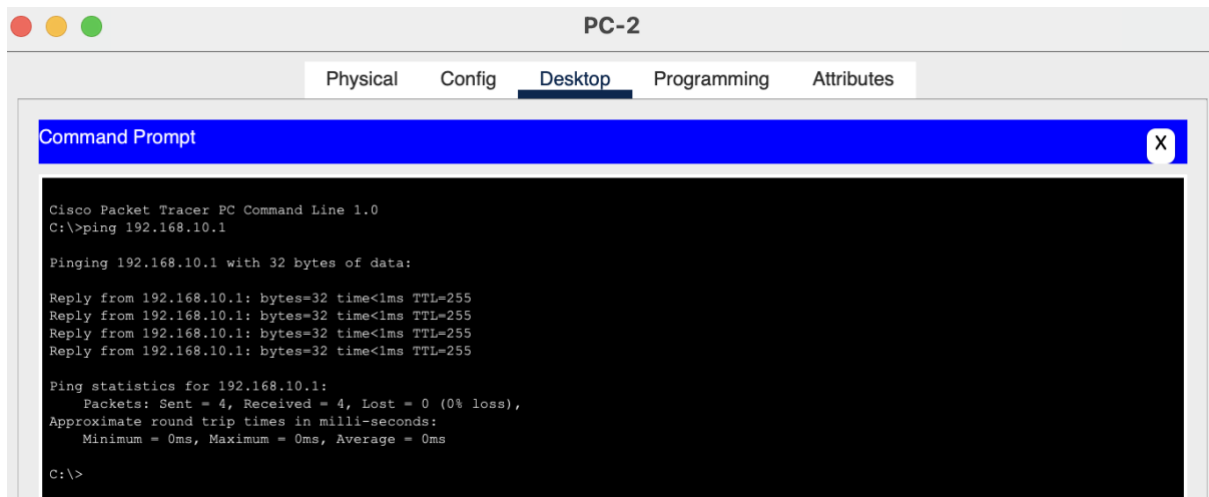
Test 1 – DHCP:



Result: Confirms correct subnet and IP Address.

Test 2 – Gateway:

ping 192.168.10.1



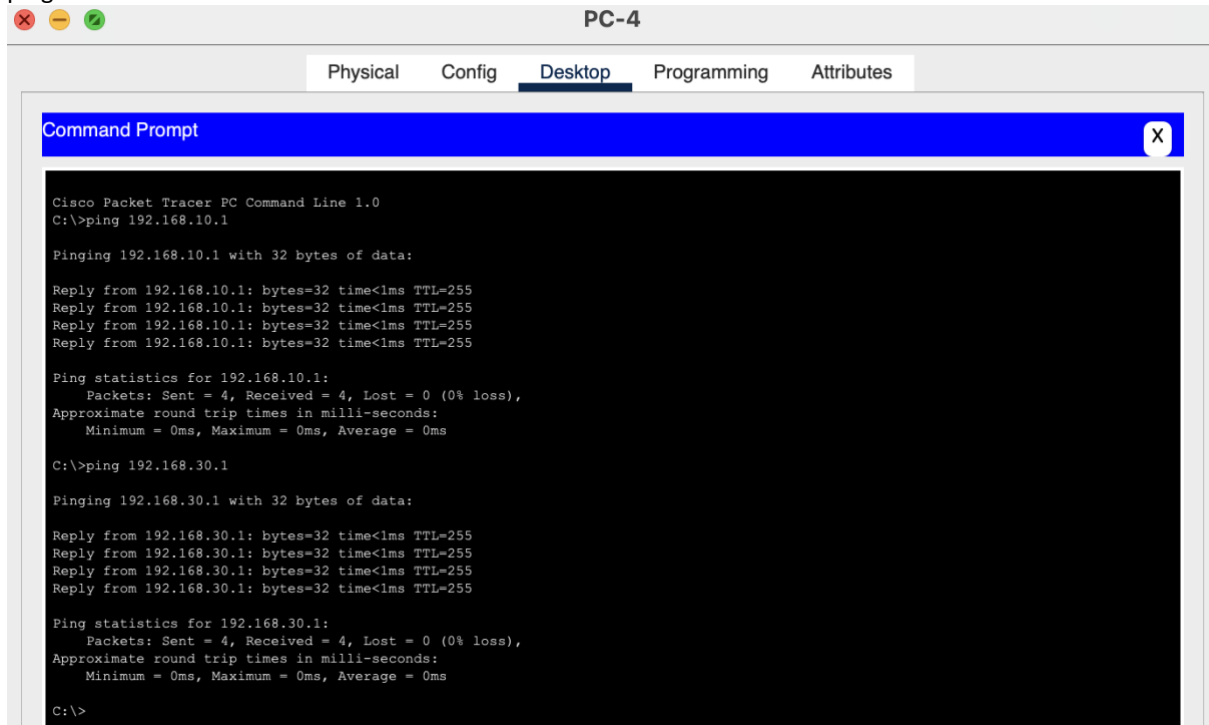
Result: All packets sent to default gateway with no issues.

Test 3 – Inter-VLAN Routing:

From VLAN 20 (SALES) PC:

ping 192.168.10.1

ping 192.168.30.1

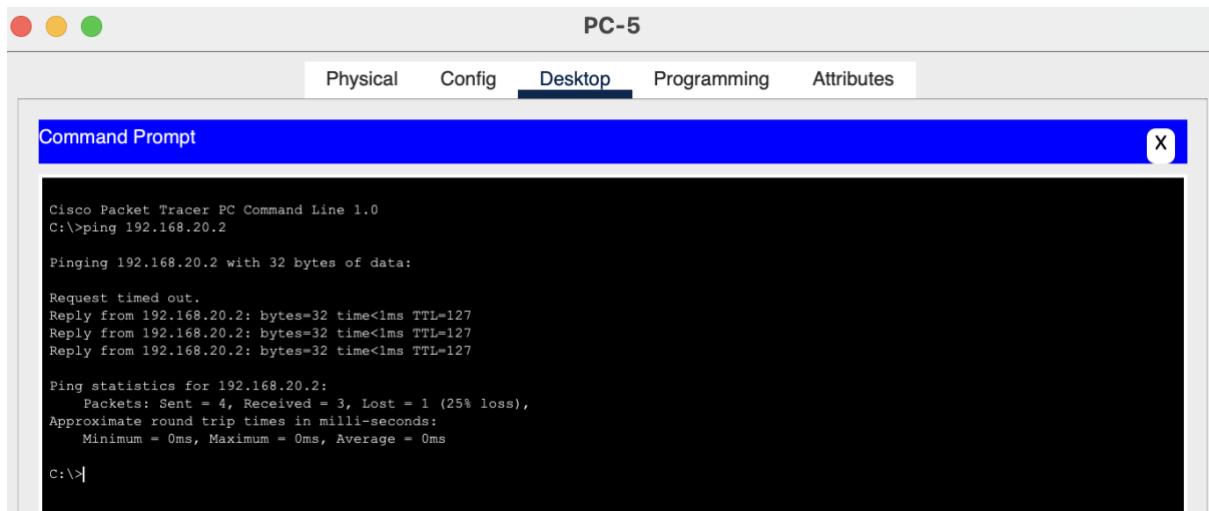


Result: PCs in each VLAN are able to ping the other default gateways

Test 4 – End-to-End Connectivity

PC in VLAN 30:

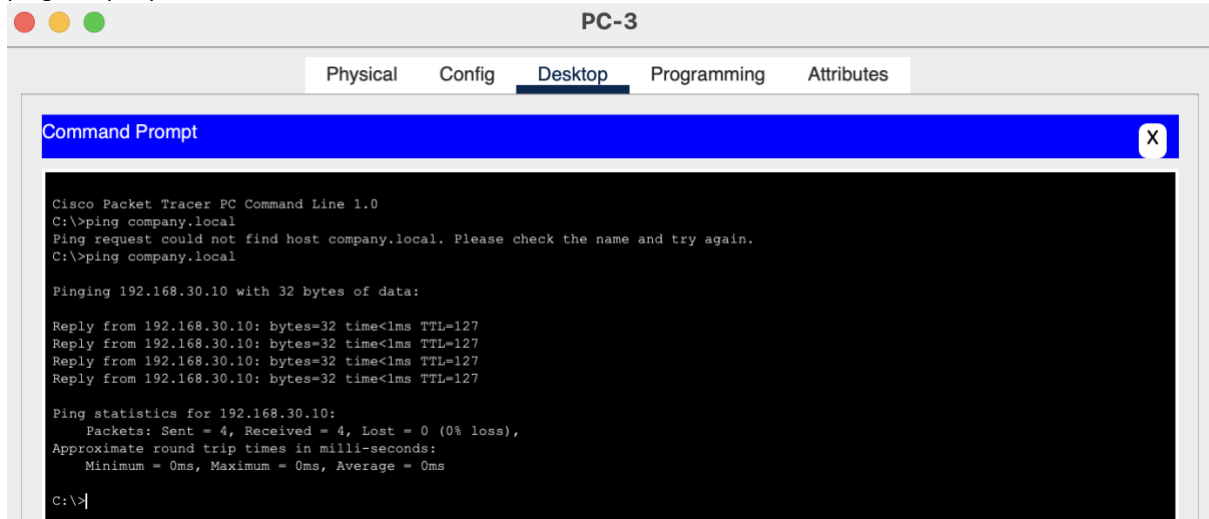
ping <IP of PC in VLAN 20>



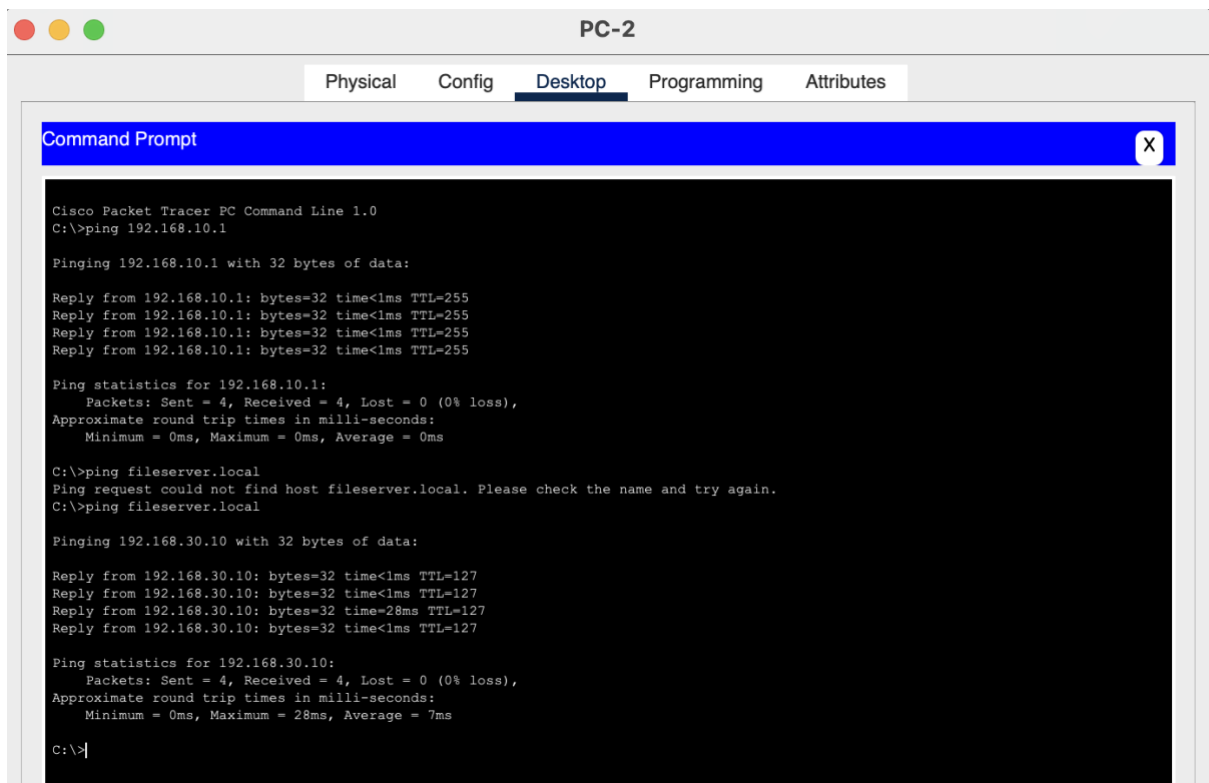
Result: First request timed out but all other packets sent by PC-5 were received by PC-3.

Test 5 – DNS

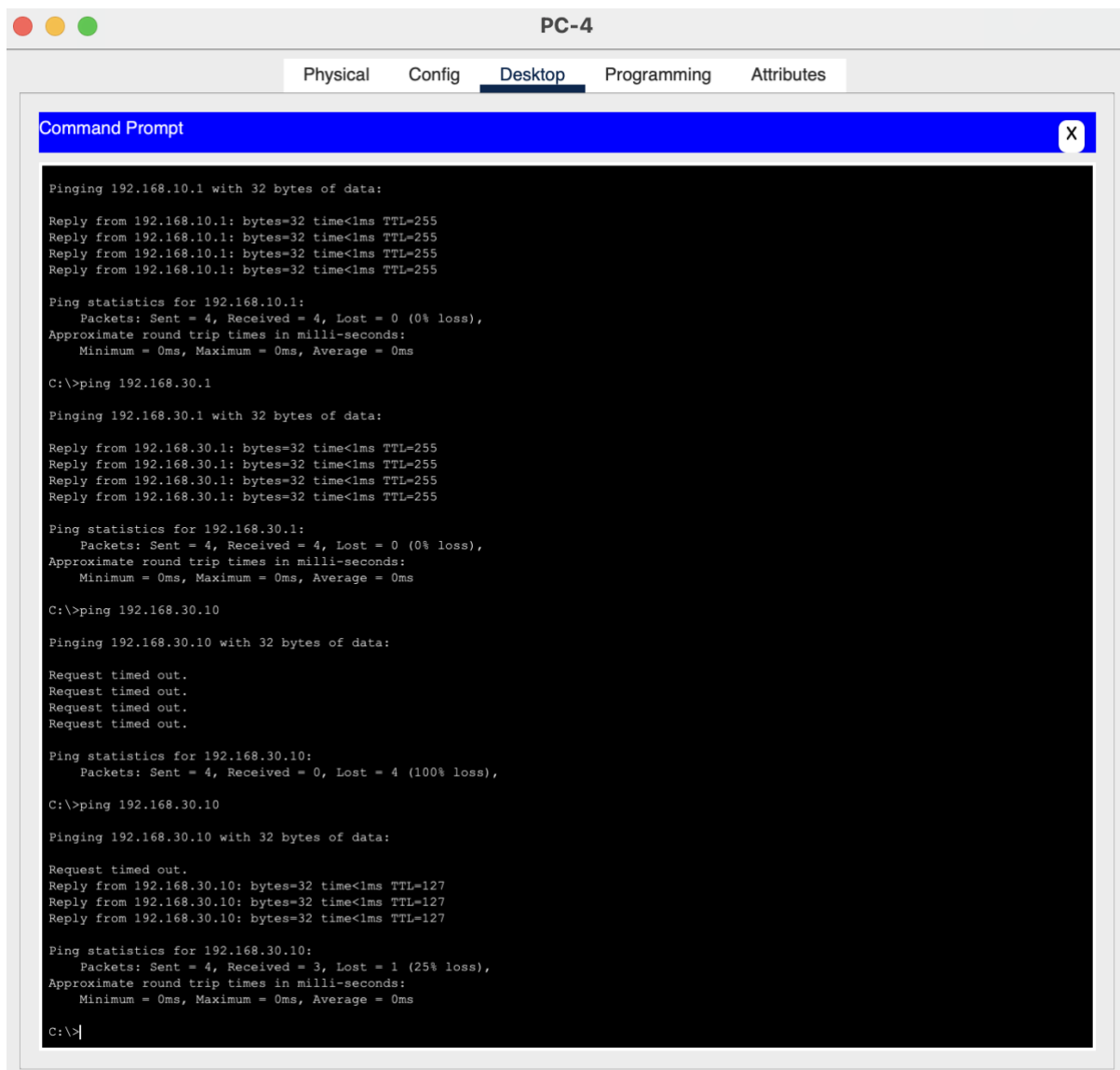
ping company.local



ping fileserver.local



ping 192.168.30.10



The screenshot shows a window titled "PC-4" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of several ping commands. The first ping is to 192.168.10.1, which succeeds with 0% loss. The second ping is to 192.168.30.1, which also succeeds with 0% loss. The third ping is to 192.168.30.10, which fails with 100% loss. The fourth ping is to 192.168.30.10, which succeeds with 25% loss.

```
Command Prompt

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Result: The DNS resolves to 192.168.30.10 and multiple PC's are able to ping the DNS Server

10. Conclusion

The network was successfully configured using VLAN segmentation, trunking, router-on-a-stick inter-VLAN routing, DHCP for dynamic host addressing, and DNS services for hostname resolution. Connectivity tests confirmed successful communication between VLANs and proper DNS functionality.